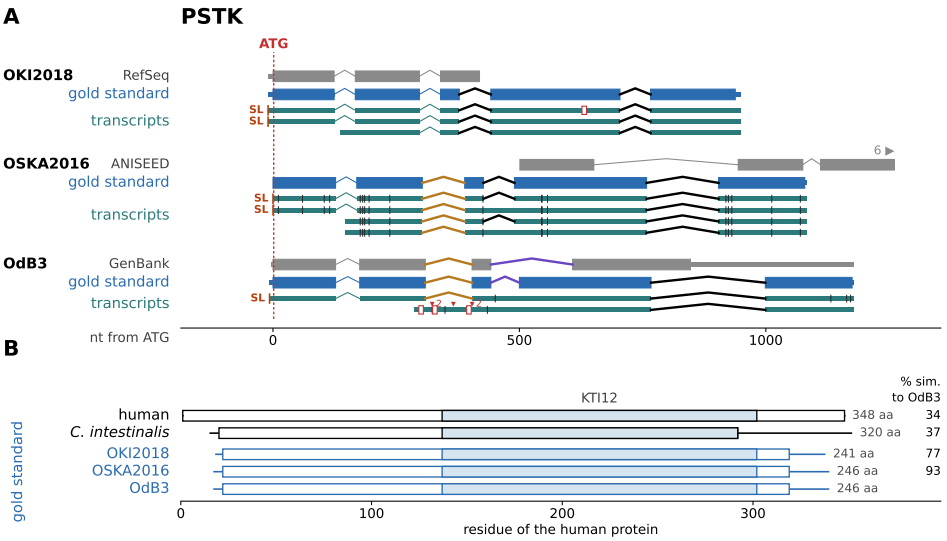


OKI2018 RefSeq XP_081642556.1: first three exons only; reads on into the third intron (110 codons, 98 of them as in the gold standard). Gold standard: third and fourth introns with splice sites outside the intron classes (TG-AG, AG-AA), as in the three supporting transcripts. Transcripts (top to bottom): TRINITY_DN16768_c0_g1_i1 (SL), TRINITY_DN16768_c0_g1_i3 (SL), TRINITY_DN16768_c0_g1_i4.

OSKA2016 ANISEED S10.g09825.01.t: starts in the fourth exon and continues past the stop codon into seven exons downstream (604 codons). Gold standard: third and fourth introns with splice sites outside the intron classes (AA-AG, TT-GT), as in the supporting transcripts; the third intron is retained in two of them. Transcripts (top to bottom): comp20361_c0_seq1 (SL), comp20361_c0_seq5 (SL), comp20361_c0_seq6, comp20361_c0_seq4.

OdB3 GenBank CBY23135.1: same first three exons; other first site of the fourth exon, and no last exon (178 codons). Gold standard: exon chain set by the curator: the GenBank exons 1 to 3, the acceptor of the third intron (GA-AG; retained in both ESTs) from the prediction, and the last intron of the ESTs (AT-AG). Transcripts (top to bottom): KT0AAA259YN02RM1 (SL), KT0AAA109YN17RM1. Also starting with the SL, not drawn: FP831546, FP830122.

proteins human Q8IV42; C. intestinalis XP_002120985.2.

C Equivalent placements of introns

Third intron

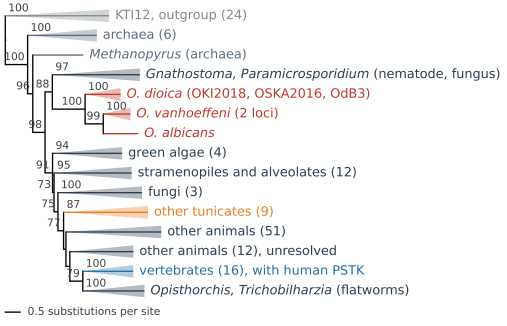
```
OKI2018      cgaataaTGCAGGGT … TTATCAGtctttatt  TG-AG GC-GT
KUM-M3-7f    cgaataaTGCAGGGT … TTATCAGtctttatt  TG-AG GC-GT
               R  I              I          F  I
OSKA2016     gtcgaatAGAAGGGT … TCAACGagttttcgt  AG-CG GA-GA AA-AG
AOM-5-5f     gtcgaatAGAAGGGT … TCAACGagttttcgt  AG-CG GA-GA AA-AG
               R                 I          V  F
OdB3         gtcgaatAGGAGGGT … TCAACGagttttcgt  AG-CG GG-GA GA-AG
Bar2_p4      gtcgaatAGGAGGGT … TCAACGagttttcgt  AG-CG GG-GA GA-AG
               R                 I          V  F
```

Fourth intron

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OKI2018      agcacttAGTTTGGG … TTGAAagtagaaatc  AG-AA GT-AA TT-AG TT-GT
KUM-M3-7f    agcacttAGTTTGGG … TTGAAagtagaaatc  AG-AA GT-AA TT-AG TT-GT
               H                 L          V  E  I
OSKA2016     aacatttAGTTTGGG … GTGACagtagaaaat  AG-AC GT-CA TT-AG TT-GT
AOM-5-5f     aacatttAGTTTGGG … ATGACagtagaaaat  AG-AC GT-CA TT-AG TT-GT
               H                 L          V  E  N
OdB3         aacatttAGATCGGG … TGTGTCagtggaaaa  AG-TC GA-CA AT-AG
Bar2_p4      aacatttAGATCGGG … TGTGTCagtggaaaa  AG-TC GA-CA AT-AG
               H                 L          V  E
```

Gene strand, cased for the leftmost placement: lowercase grey, exon; uppercase, intron. Shaded: bases found at both ends, across which the intron can be moved without changing the spliced sequence. Letters below: amino acids of the complete exon codons and, under the ellipsis, of the codon split by the intron, once per species (the test assembly translates the same). Right: dinucleotides of each placement, from left to right; blue: placement on an AG acceptor (phase 1); ▲: placement named by the gold-standard script. Rows in italics: test assemblies (other individuals of the same species), exons carried from the gold standard.

D Simplified tree of PSTK, more eukaryotes and archaea (PMSF)



Nodes supported by less than 50% collapsed; ultrafast bootstrap supports of 70% or more shown. Wedges: clades of sequences of one group, from their ancestor to the deepest sequence; “unresolved”: sequences of the other groups in one polytomy, drawn as one tip. Sequences added to the tree of Figure TREES: see Methods.

exon: coding (thick), UTR (thin) · public model · gold-standard model · coding sequence in another frame

start codon of the gold-standard model (x axis: nucleotides from it, gene orientation) · ◂n number of exons beyond the drawn region

a b public models of the same row, as named in the paragraph

transcript aligned to the assembly (exons) · bases of the transcript missing from the assembly (number when more than one)

bases of the assembly missing from the transcript · mismatches · SL transcript that starts with the spliced leader

intron splice sites (drawn length unchanged): GT-AG (colour of the model) · GA-AG · GG-AG · other site

Pfam domain of the first reference · residues not aligned to the first reference

92 % similar residues to the OdB3 protein (BLOSUM62 > 0; global alignment, aligned part) · human: UniProt; Ciona: RefSeq